Road to Super Artificial Intelligence: Is Artificial Intelligence Complete without Artificial Emotions

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Abstract—**For years now Artificial Intelligence has been occupying time and research efforts of vast number of computer scientists. The field of Artificial Intelligence was created at a conference on Dartmouth College in 1956. At starting point brave scientists were claiming that machines as intelligent as humans will emerge in no more than a decade. Today it is year 2015 and still no machine that can be described as fully intelligent. Methodology used for deciding if a machine is intelligent or not is Turing test, and to this day no machine has ever passed it (the debate on the fact whether *Eugene* system has or has not passed is still 'raging' on). It is clear that initial prognosis was overly optimistic. What is a central point that made artificial intelligence one of the hardest problems in the modern computer science? Is there something missing in the initial concept?**

I. Introduction

Aim of this paper is Artificial Intelligence, with restriction that its goal is not to define/re-define Artificial intelligence as a discipline, but to give a detailed description of Artificial Intelligence as a concept, meta-level object, model of behaviour, cognition and emotion of artificial life. Exhaustive definition of artificial intelligence as a concept is proposed in the section II. Reader should take content of this paper as new angle on the topic and not as an attempt to redefine skyline of modern Artificial Intelligence field.

In the section III emotions as a part of intelligent beings are introduced and models, such as Plutchik's wheel of emotions and high/low road to amygdala, are considered and explained. At this point correlation between intelligence and emotions is established and concepts of conscious and sub-conscious behaviour are cast into domain of intelligence as biological organisms observe it.

Borrowing Marvin Minsky's modelling techniques notion of self-awareness and consciousness is made as a central point of the discus. Later this discus moves its focus on the topic of artificial evolution, on the possibility that artificial intelligence evolve itself beyond initial design. This question has already raised fears in eminent experts such as Bill Gates and Stephen Hawkins. In the scope of this paper the focus is not on the doomsday prediction, and fear of consequences of artificial intelligence surpassing humans. Goal is to provide critique on models of artificial intelligence that could possible observe this evolutionary quality.

In the terminus of the paper a model of artificial intelligence is proposed and described in detail, as long with its comprising parts. All these models are used as a tool to depict to the reader hierarchical structure of intelligence as a concept. This model borrows concepts described throughout the paper, such as emotions, Plutchick's wheel of emotions, creativity, motivation, evolution, etc. This model is extremely complex and it should not be observed as strict instructions how to build artificial intelligence but just as one attractive possibility.

II. What is Artificial Intelligence

To be able to define *Artificial Intelligence* there should be a consensus on definition of *Intelligence* (what humans consider intelligence to be). Cambridge dictionary defines *Intelligence* as follows [1]:

"The ability to learn, to understand, and to make judgements or to have opinions that are based on reason"

Encyclopedia Britannica defines Human Intelligence as [2]:

"Mental quality that consists of the abilities to learn from experience, adapt to new situations, understand and handle abstract concepts, and use knowledge to manipulate one's environment"

The second definition seems more precise and gives clearer description; also it includes the notion of *mental quality* this is highly correlated to the notion of *qualia*, used by many researchers in the field of *Artificial Intelligence*. Both of these definitions, although precise and clear, raise the question of completeness. Is this how humans perceive their own intelligence? Is there something missing in these definitions?

From the earliest periods of history humans tend to make abstract models of the world around them. Trying to model the reality with the simplest model possible, often deliberately leaving out some aspects of the real world. So, are the previous definitions, such abstract models of the real biological system inhabited in human brains (one could say also in human souls, but this could lead to debate on existence of this notion in the real world)?

These models of intelligence have served their purpose well in the past; even so there are limits in their descriptive power. There are some qualities that humans observe on daily basis; yet they are omitted from previous definitions. To increase descriptive power of the model one can add missing attributes to the system and therefore increase the power but also the complexity of the system.

One of the missing attributes is the so-called *Emotional Intelligence*, defined by [10]:

"Ability to understand and control your own feelings, and

to understand the feelings of others and react to them in a suitable way"

With this attribute added to previous description, intelligence can be observed as a system that at the same time is able to operate on a cognitive and emotional level. Even though newly formed system seems more descriptive and highly complex it still lacks the depth. If the definition is self-applied then the notion of self-awareness is obtained. With cognitive intelligence, emotional intelligence and self-awareness the complete definition is formulated:

The organism (or in technical terms, the system) can be considered intelligent if it observes:

1. Cognitive intelligence: mental quality that consists of the abilities to learn from experience, adapts to new situations, understand and handle abstract concepts, and use knowledge to manipulate one's environment
2. Emotional intelligence: ability to understand and control your own feelings, and to understand the feelings of others and react to them in a suitable way
3. Self-Awareness: ability to create a model of one's self both on cognitive and emotional level, as well as ability to integrate this model inside of the greater domain, the model of the word one has, and correlates self-model with models of others inside the world model

Now that the definition of intelligence is established it is fairly simple to define *Artificial Intelligence* (later referred as *AI*) as mechanical counterpart to Biological Intelligence (later referred as *BI*):

AI is quality of any artificial system (not necessarily only computers) that can exhibit all three components of BI:

1. Cognitive intelligence
2. Emotional intelligence
3. Self-Awareness

After all this being said one can argue that emotional part of intelligence can be counterproductive in some situations, as all humans have experienced on various occasions inability to act rationally due to emotional mayhem. On the other hand if this important part of intelligence is omitted, *AI* can never exhibit all qualities of *BI*.

Also one can argue that Self-Awareness (or Self-consciousness) is not mandatory part of the AI. This opens question of system being able to improve itself if it is not aware of what it is, thus no self-awareness implies no self-improvement, and as previously said this is one of the aspects of cognitive intelligence.

III. WHAT EMOTIONS ARE

Definition proposed by James-Lange takes into account external event that causes physiological changes in the body followed by arousal [3, 9]. These physiological changes can be seen as signals that human body processes and interprets them as emotions. This black-box description can be understood as filter inside a human brain through which chemical stimuli are passing and the outputs are emotions.

This definition is flawed to some extent. The whole notion of processing the stimuli is not clearly explained. Also the context of the situation is missing. For human body to fully appreciate emotions it is not enough just to describe them as chemical reaction, the situation in which arousal is produced plays significant part as well. Same feeling of raised heart beat can be observed in two completely different situations:

1. Benjie jumping activity: which will be interpreted as pleasant experience (assuming that one who is experiencing this activity is favourable of extreme sports)
2. Being in vicinity of the ongoing robbery: which will be interpreted as highly traumatical experience

Therefore the context of experiencing emotions plays major role in intelligent system interpretation of the stimuli. This observation leads to biological solution in deciphering emotions. Part of the brain called amygdala is responsible for this activity. Brain research lead to discovery of two emotion pathways in the human brain; both of them go through amygdala [4]. Shown in the figure 3.1.

The low road to the amygdala is more reflexive way of dealing with emotions, the pathway is shorter and it doesn't include reasoning functionality of the cortex. The high road to the amygdala includes more elaborate functional activities in the brain; it represents combination of cognitive and emotional reasoning.

Up until now emotions were defined at primal, basic level. Emotions are not just reflex chemical reactions to outside stimuli; emotions can be self-inflicted and recurrent phenomenon. According to Plutchik's Wheel of Emotions (1980) there is a hierarchical structure to be observed in

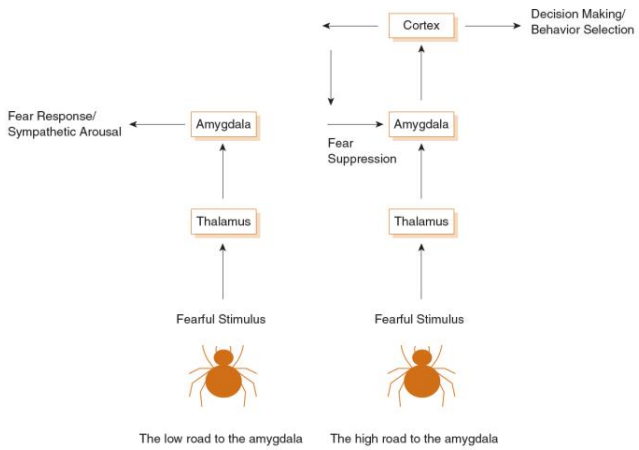

Figure 3.1. Stimulus pathways through amygdala

emotions; and from this hierarchical structure combinatorial and equipoised qualities of emotions can be deduced [5].

This hierarchy of emotions implies a fruitful field of signals and stimuli that intelligent organism (or a system) can use to

prioritize internal and/or external activities both on conscious and sub-conscious level. One could observe that current artificial systems lack consciousness, and therefore it is quite absurd to include hierarchical split between conscious and sub-conscious intelligence in a system/organism that has no

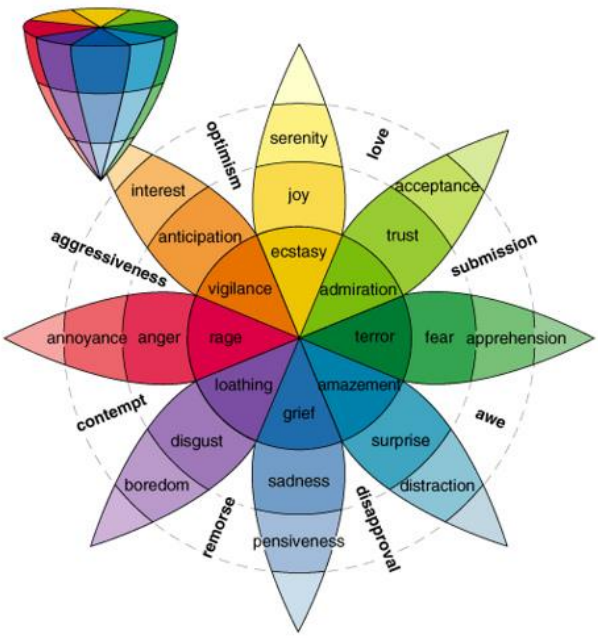

Figure 3.2. Plutchik's Wheel of Emotions

consciousness. Never the less, including emotions inside the definition of intelligence implies a whole new level of primal instincts that the intelligent organism/system should exhibit. Therefore split between conscious and sub-conscious level is logical decision in design of superior intelligent system with emotional aspects that penetrate this barrier.

Conscious Level: • Cognitive abilities • Emotion generation • Emotion interpretation • External stimuli	Emotions
Sub-Conscious Level: • Emotion generation • Emotion interpretation • Internal stimuli • Motivation • Denial • Creativity	

Figure 3.3. Conscious and Sub-Conscious

After defining emotions by combining psychological and biological definitions, one can't help by wonder why this model is rarely used in practise in the modern AI systems. To reinforce this observation, one should take into consideration reinforcement learning systems (later referred as *RL*). *RL* systems rely on 'learning' from environment through reward/punishment stimuli. Both reward and punishment in biological organism would induce emotions of happiness and sadness (possibly anger, or even motivation) respectively. The issue with *RL* systems arise from the fact that *RL* system is not self-aware, therefore it doesn't observe conscious and sub-conscious qualities of biological counterparts. One can argue that since *RL* system has no self-awareness and observes no internal emotions, *RL* system can't be learning in the full sense of this word. And this would be right conclusion to be made; *RL* systems, even though considered to be *AI* systems, are nothing more than sophisticated equation solvers (Bellman's equations [6]) based on iterating trough all possible actions system could take in all possible situations. This quality of *RL* system can be observed as learning, but only a limited type of learning. *RL* systems learn how to improve their parameters, but never learn about their internal structure and never improve upon their structural attributes. Therefor this example can be considered as only technical and limited type of intelligence and learning abilities.

IV. Self-Model

After cognitive and emotional parts of intelligence are explored in more details, next logical step is to tackle the concept of self-awareness. This concept was introduced by Marvin Minsky in 1968, at this time he wrote a paper in which he proposed a concept of self-model.

"To an observer B, an object A* is a model of an object A to the extent that B can use A* to answer questions about A that are of interest to B. If A is the world itself, questions for A are experiments. A* is a good model of A, in B's view, to the extent that A*'s answers agree with those of A, on the whole, w.r.t. to the questions important to B. When a man M answers questions about the world, then (taking on ourselves the role of B) we attribute this ability to some internal mechanism W* inside M." [7].

This definition can seem cryptic to one not too familiar with formal logic and modelling techniques; what is essential to be taken from this definition is the fact that intelligent organism/system will build approximate models of objects and world around itself and will further use these models to make cognitive and emotional responses/decisions (precise models are infeasible and/or undesirable, since even the simple phenomena of feather falling from 1 meter height would include complex consideration of temperature, atmospheric pressure, humidity, latitude and longitude of location, etc.).

Another, extremely important, point to be taken from this definition is that W* (world model) includes *M** (self-model) [8]. This notion is recurrent and can go indefinitely deep, how *M* sees the world *W* is defined in *W**, but how *M* sees itself inside of the world is contained in $M^* \in W^*$, further what *M*

observes of its own role inside of the world would be contained in $M^{**} \in W^{**}$, $W^{**} \in M^{*}$, this line of thought can be followed even further. For human it would be unnatural to go deeper that 2-3 levels; still this doesn't exclude existence of infinite structure of self-including models.

Based on this discussion one can try to provide formal description of how *AI* system could model the world *W*. Taking into account that this is writers observation, ergo part of the *W** defined by the writer, *AI* should be regarded as $AI^* \in W^*$. After this being established, next step would be defining desirable expression power of the *AI**. *AI* should be as close as possible to *BI*, which implies that *AI** should be as close as possible to *BI**. *BI** is nothing more than the definition of *BI* proposed in the section II accordingly *AI** is the definition of *AI* proposed in the same section. Both of these definitions include self-awareness part of intelligence:

Ability to create a model of one's self both on cognitive and emotional level, as well as ability to integrate this model inside of the greater domain, the model of the word one has, and correlates self-model with models of others inside the world model

By breaking down this definition the following pieces are obtained: world model *W***, self-model *AI***, *CI*** (cognitive part) and *EI*** (emotional part) which together comprise *AI***. What should be noticed is that the numbers of stars in superscripts of models are correct, since the starting point was *AI**, the model writer has of *AI*. It would be illogical to claim that *AI** is what *AI* actually is to its full extent; obviously it is just writer's approximate.

Putting it all together leads to explaining notion of self-awareness and artificial self-awareness, *AI* that exhibits self-awareness. Consider a man *M* (*M* is an embodiment of *BI**) that has a model of the world *W**. When *M* is asked "What are you?", *M* would answer with *M**, which means that *M* is aware of its own existence inside world and has a model of itself [7]. When *M* is asked "How your mind works?", *M* would answer with *BI***, which means that *M* is able to explain, to some extent, the way it understands its own mind. One can observe that *M*** includes *BI***, being able to understand its own mind implies being able to understand how it understands.

Now consider intelligent computer *IC*, it is expected that *IC* exhibits *AI**. If *IC* is exhibiting attributes of *AI** then it has a model of the world *W** and inside of *W** it contains *AI***. Ergo if *IC* has model of *AI*** it understands how it understands and it is self-aware at the same time. Also as stated above it is clear that *AI*** is composed of *CI*** and *EI***. It is clear that *AI** contains all needed components of intelligence as defined in section II.

At this point one can argue that these models are approximation and that *IC* will never be as smart as *M*. This argument is easily disproved by examining that *AI** and *BI** are subclasses of intelligence whose only difference is the embodiment, for *AI** is a machine for *BI** it is an organism, and no further difference can be stated. Other objection is that this consideration was not intended as measurement to which degree *AI** will be intelligent (obviously the writer's intention is not to compute *AI**'s IQ) but as qualitative description of *AI**; and on qualitative level w.r.t. to previous definitions *AI** and *BI** are equivalent.

V. Evolving systems, Super AI

It can be argued that no current system labeled as AI is actually intelligent. All these systems (e.g. reinforcement learning system, artificial agents, Markov Decision Processes, Natural Language Processing system, etc. [6, 8, 9]) fail to capture all concepts of intelligence. Most notable quality missing in these systems is ability to understand how they reason; the model of the world is present and the model of the system is present, with restriction that these systems change parameters inside the model but never change the model. This can be observed in the fact that there doesn't exist (at least at the moment of writing, and as far as writers knowledge goes) system that will start its execution, its existence, as model *M1* (e.g. neural network) and evolve itself to model *M2* (e.g. artificial agent with expert knowledge in stocks).

If this quality was observed in any state of the art system that would directly lead to embodiment of superintelligent *AI* (later referred as *SAI*). If a system is Intelligent and able to evolve, when would this evolution process end; what would be the outcome of the process? *SAI* by definition is an *AI* that surpasses any human in every aspect of intelligent behavior [1]. Taken in the context of this paper, *SAI* is the intelligent system whose model of itself *SAI** is more expressive than any *M** of any *M*, or more precisely any *BI** of any *BI*. This will lead to conclusion that *SAI*'s *W** could incorporate better *BI#* than any *BI** of any *BI*. Putted in simple terms, *SAI* would understand organisms (including humans) better than they understand themselves.

This concept may sound ludicrous, but it incorporates same reasoning as of that humans better understands rats than the rats understand themselves (even though this is an assumption). So this hyper intelligent system would be highly superior to humans in cognitive aspect for sure. What about emotional, or self-awareness aspects of intelligence? The initial definition of *SAI* states that *SAI* surpasses humans in every aspect of intelligent behavior. Again after consulting previously stated formalism it can be easily noted that *SAI* will have a model denoted by *SAI** (which corresponds to *AI***) and this model will include *SCI** and *SEI** which represent super cognitive intelligence and super emotional intelligence, respectively. *SAI* would also have self-awareness part of intelligence, otherwise it couldn't possibly be considered as intelligent and superior to humans. This leads to conclusion that *SAI* if self-aware at the same time should be aware of its superiority over *BI*; and *AI* as its previous state in evolution.

If *SAI* is superior to humans and incorporates the best possible model of *M** and *M** is better model of *M* than any *M#* any *M* has of itself, this implies that *SAI* couldn't have been implemented by humans (assuming that *SAI* is not implemented by some extraterrestrial species that is more intelligent than humans). Which can mean only one thing: *SAI* is a product of *AI* system evolving itself beyond initial design.

For *AI* system to able to evolve *AI* should have clear

understanding of itself, this condition is intrinsic to the *AI** definition and therefore trivially satisfied. Furthermore *AI* should observe motivation to evolve, if *AI* can't see any purpose in improving itself it will never do so; as well as observing creativity, enabling AI to improve itself. Creativity might not seem at all different than cognitive ability and might seem like it adds no additional value to description of *AI*. Consider following example:

There is a world *W1* in which exist *AI* system *AI1* and man *M1* the creator of *AI1*. *AI1* is able to understand what physical force is in same level of detail as *M1*. Now assume the time of the experiment is before Isaac Newton. This implies that both *M1* and *AI1* are unaware of gravity and of Newton's Laws. Will *AI1* be able to infer gravity by only using cognitive reasoning? Or *AI1* would need some outside 'push' to achieve this breakthrough?

At this stage of discus creativity can be observed as interrupt signal of the brain, an emotion that is triggered by sub-conscious part of the human intelligence. This interrupt causes human (in the context of previous example Isaac Newton) to switch the attention toward the phenomenon of which was not aware and then after the context switch use its cognitive abilities to infer model of the phenomenon only if reasonable motivation is present. Therefore combination of creativity and motivation is need by *AI1* as well to be able to invent.

Returning to the concept of *SAI*, *SAI* is a product of *AI* evolving beyond its initial design. For *AI* to be able to reinvent itself it must observe creativity and motivation, that are considered to be part of *EI** on sub-conscious level. *SAI* may or may not decide later in its evolution to break the split between consciousness and sub-consciousness; no assumptions on decisions of *SAI* will be made at this point since it will bring no additional value to the argument.

VI. Evolutionary AI system model

Now that the concept of *SAI* and *AI* are well established, as well as the notion of artificial evolution, model of the *AI* system can be proposed. This model should be observed on highly conceptual level as there are no clear indications that implementation of such model is feasible at the time of the writing.

As one would have expected *AI** will mimic as much as possible behavior and qualities of biological counterpart. Choice of biological organism that will be mimicked by *AI** logically falls down to human, as this is the most intelligent organism known and whose intellectual models are described. One can argue that existence of higher extraterrestrial intelligence is highly likely, but one can't base critique on plausible unproven notions. The description used in this paper is the psychological/biological description of *BI* described in section III. In this description three sub-models are observed:

- Sub-model designated for conscious behavior and reasoning – further denoted as *CB**
- Sub-model designated for sub-conscious behavior and reasoning – further denoted as *SCB**
- Sub-model designated for emotional transfer of information inside of the *AI** model – further denoted as *EM**

These three models are able to interact with each other and (as depicted in fig. 3.3.) for interaction they will use *EM**. To make this point clearer *EM** should be presented on level beyond just stating its name and saying it is a model of emotions inside of a model *AI**. Model *EM** can be observed as model of Plutchik's wheel of emotions. Any other sophisticated psychological model of emotions inside intelligent organism is as good choice as this one. With one restriction: a model should capture qualities of real object as much as possible, even though precision is not main attribute of models inside of Minsky's world. Plutchik's wheel of emotions exhibits combinatorial and equipoised qualities in its description; this makes it highly attractive choice. It does not just simply list emotions it also describes interrelations of emotions: how they oppose one another, how they can be combined, and hierarchy inside a family of emotions.

After this being said defining *EM** is straightforward. *EM** is composed of 8 emotion family models EF_i^*, each of which corresponds to one of the 8 slices in the wheel. Each of EF_i^* is composed of 3 emotion models E_i^*. Relations between EF_i^*s are defined as described in the wheel, each EF_i^* has one directly opposing emotion family EF_j^* and two partially opposing families EF_{k1}^* and EF_{k2}^*; on the other hand each EF_i^* is related to EF_{g1}^* and EF_{g2}^* with whom it can be combined to produce complex emotions $ECMP_i^*$. Complex emotions, as depicted in the wheel, have no family (or one can place them in the 9th family, this bares no difference in the construction of the model whatsoever).

Secondly, the sub-model *CB** is defined so it captures following qualities of *AI**:

- Cognitive abilities
- Emotion generation
- Emotion interpretation
- External stimuli detection

Cognitive abilities can be regarded as sub-model of their own, denoted as *CA**. *CA** is comprised of all reasoning functionalities and algorithms incorporated in the proposed model. Each of these functionalities should have a model of its own, denoted as RA_i^*. One should observe importance of this statement, since *AI** is regarded as a model that is able to evolve, this implies that higher the granularity of the model higher the flexibility of the evolution. *AI** model at this level of granularity and complexity can be regarded as meta-level description of *AI*'s *DNA* (again at highly abstract level) and from biological world is known that more information presented in the *DNA* more advanced organism it is (higher level of evolution).

Emotional interpretation is a bit trickier to be defined as it is observed both on conscious and sub-conscious level. Mitigation to this complication is taken from amygdala roads analogy in *BI*, described in the section III. Emotional interpretation on conscious level is the model of high road to amygdala whereas emotional interpretation on sub-conscious level is the model of low road to amygdala. One can

understand this as fact that some emotions don't have need of cognitive reasoning. These two models are denoted both as *EI**, one of them belonging to *CB** and the other to *SCB**. For more detailed description of these sub-models consult section III.

External stimuli detection model denoted as *ESD**, is composed of models of sensors that are able to perceive outside world. Each of these sensors is modeled with its own model S_i^*. For one to understand what S_i^* is, one can consult real world example of pressure sensor, light sensor, face recognition system, etc. All this real world systems have their own models and are implemented in accordance with these models. It is noticeable that time and resources needed to collect and devise all this models could be extremely high, still this doesn't imply impossibility of such endeavor.

Now that *CB** is extensively defined and described next and final step is describing *SCB** model. This model is crucial for the idea of evolutionary, therefore self-improving *AI*. Starting point in definition of *SCB** is to observe its intrinsic qualities. By consulting the description of *BI* from section III following qualities of *SCB** are observed:

- Emotion generation
- Emotion interpretation
- Internal stimuli detection
- Motivation
- Denial
- Creativity

Emotion interpretation is already explained on both conscious and sub-conscious level; therefore this explanation won't be repeated. Internal stimuli detection as sub-model can be observed at equivalent line of thought as when external stimuli sub-model was defined. Only difference is that of the origin of the stimuli which on a conceptual level bares no difference.

Emotion generation sub-model, later denoted as *EG**, is the model that, as well as *EI**, is present both in *CB** and *SCB**. Together with *EI** *EG** allows *AI** to exhibit communication between *CB** and *SCB**. *EG** can be observed as triggering model, this model would take as input the output of external/internal stimuli sub-models and will instruct to *EM** which E_i^* to duplicate and send to the *CB*/SCB** depending on the current flow of the emotions. At this point it becomes clear that concepts of Motivation, Denial and Creativity will reside inside of the *EG** model. Each of this concept will be embodied inside its own sub-models as parts of *EG**, denoted as *M**, *D** and *C** respectively.

These three models hold the key to the AI evolution. Whenever an emotion E_i^* is observed it will be firstly sent to *M**, *D** and *C**. *M** is the model that will understand when this emotion was earlier observed and if it had produced desired outputs, if so *M** can increase the intensity of the emotion (swap it with one higher in the Plutchik's wheel). *D** on the other hand will be able to block undesired emotions, emotions whose exhibition have produced harmful outcomes to the *AI** in the past. *C** behaves in slightly different way. *C** will recreate emotions when not present in a context similar to the ones that were observed in the past. In addition it will create only emotions that produced positive outcomes in previous similar contexts.

At this point only thing missing from initial definition of *AI** from section II is self-awareness. Self-awareness is achieved by *AI** having exact model of itself, denoted *AI*** (since this is artificial system it is possible to hold exact model of itself, the model on which its implementation was based on). This being said evolution is defined inside of the *EV** model that belongs to *AI**. *EV** is in charge of modifying *AI*** and swapping any sub-model *SM** belonging to *AI** with *SM*** belonging to *AI***. *EV** model should be observed as pass-through filter of emotions, it will detect all emotions inside *AI** but will not change them or affect their flow. This way functionality of *EV** and those of *EM** and *CA** are in no way inter-blocking so the *AI** can reason and improve itself at same time. The only moment in which a block can be observed is at the moment when *EV** swaps parts of *AI** with those of *AI***, but this argument doesn't affect goal of the discussion.

VII. Conclusions

At this point one should be convinced that the only way of achieving *SAI* is by *AI* reinventing itself. Put in more down to earth language, one can't expect that human intelligence is created by insects, therefore one can't claim that human can create intelligence greater than its own, at least not directly.

After reading section VI one should notice that crucial points in creativity and motivation are emotions and if we remove *EV** and *EM**, *AI** will never try to improve itself. One can argue importance of the emotions by claiming that one can devise an algorithm that will improve *AI* over and over. If one could device such an algorithm, this algorithm by itself would be an embodiment of *SAI*, since could increase its intelligence without limit. At this point one can say that this argument makes whole discussion futile; luckily this could be true only if the intelligence is observed as solely cognitive intelligence. This hypothetic *AI* system would only include *SCI* but not *SEI*; by definition *SAI* surpasses humans in every aspect of intelligent behavior and this includes also *EI*.

Another and final possible objection to the proposed model *AI** can be raised by asking following question: "Is *AI** really intelligent?". To answer this question one should firstly ask: "Does *AI** observe all aspects of intelligence?". The answer is: "Yes.". *AI** can perform cognitive reasoning due to *CA** (some of these activities far better than humans), *AI** can observe and understand emotions due to E_i^* models (these can be really detailed and precise models of emotions; no requirement on which implementation should be chosen is stated) and due to *AI*** *AI** is self-aware. On top of observing all these concepts of intelligence *AI** is able of self-evolvement due to *EV**, this can lead *AI** to evolve into *SAI**. *SAI** can be achieved only if improvement is unbounded. Only possible bound of improvement could be *BI**, which is a model of human, but even in initial design *AI** outperforms human in some cognitive activities, trivially proven by including *AI* chess player sub-model inside of *CA**; therefore *AI** evolution is unbounded.

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